

CS 4780 — Machine Learning: Course Syllabus

1. Course Information

Department: Computer Science. Credits: 4. Term: Fall 2025. Meeting Time: Tuesday/Thursday, 10:00-11:15am. Instructor: Prof. A. Rossi.

2. Prerequisites

Students must complete STAT 3000 (Introduction to Statistics) or an equivalent statistics course with a grade of C or better before enrolling in CS 4780.

Students who have not completed this prerequisite must request an override from the instructor prior to the add/drop deadline, using the university's Prerequisite Override Petition. Concurrent enrollment in STAT 3000 during the same term as CS 4780 is not permitted; the statistics requirement must be completed beforehand.

3. Course Description

This course introduces the mathematical foundations and practical implementation of supervised and unsupervised machine learning methods, including linear and logistic regression, decision trees, neural networks, clustering, and dimensionality reduction. Heavy use is made of probability and statistics covered in the prerequisite course.

4. Grading Policy

Homework 30%, Midterm Exam 25%, Final Project 30%, Participation 15%. Late homework is penalized 10% per day unless a university-approved extension is granted.

5. Course Schedule

Weeks 1-2: Probability and statistics review. Weeks 3-4: Linear and logistic regression. Weeks 5-6: Decision trees and ensembles. Weeks 7-9: Neural networks. Weeks 10-11: Unsupervised learning. Weeks 12-14: Final project work and presentations.

6. Academic Integrity

All students are bound by the university's Academic Integrity Policy. Suspected violations will be referred to the Office of Academic Affairs.